

Regular Perturbation on the Group-Velocity Dispersion Parameter for Dual-Polarization Short-Reach Systems

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Abstract *The Manakov equation governs the propagation of signals in dual-polarization systems. Its solution is usually approximated by regular perturbation on the nonlinear Kerr parameter. In this paper, we propose a novel regular perturbation on the group-velocity dispersion parameter for the Manakov equation. ©2026 The Author(s)*

Introduction

Analytical models of fiber-optic propagation play an important role in the design and optimization of optical communication systems [1]. They provide insights on the interplay between chromatic dispersion and nonlinearities, enable rapid performance evaluation avoiding computationally-expensive split-step Fourier method (SSFM) simulations, and serve as building blocks for new digital signal processing (DSP) algorithms such as digital nonlinearity compensation schemes. The most widely used analytical framework is the regular perturbation (RP) on the nonlinear coefficient γ , which treats the fiber Kerr nonlinearity as a small perturbation of the linear (dispersive) propagation [2, 3]. This approach accurately describes the physical signal propagation in the linear and pseudo-nonlinear transmission regimes, regardless of the accumulated chromatic dispersion in the link.

However, the accuracy of RP with respect to γ degrades when nonlinear effects are strong, as is the case in systems operating at high launch powers, or with a small group-velocity dispersion (GVD) parameter β_2 . To address this limitation, a complementary approach was proposed in [4, 5]: the *RP on the GVD parameter* β_2 , which treats GVD as a small perturbation and expresses the solution as a power series in β_2 . This approach works particularly well in regimes where nonlinear effects are beyond the reach of the RP on γ and where GVD is limited. Examples include transmission in the O-band (near the zero-dispersion wavelength) or in non-zero dispersion-shifted fibers at moderately high powers. In [5], the *RP on β_2* was developed for single-polarization systems and validated using the SSFM, demonstrating superior accuracy over RP on γ in the weak-dispersion regime but showing its own limitations at high symbol rates.

A key application area for *RP on β_2* is the emerging O-band coherent data-center intercon-

nect (DCI) segment, an operating scenario of growing relevance [6]. Standard single-mode fiber (SMF) has near-zero dispersion at 1310 nm. In this regime, the conventional RP on γ rapidly loses accuracy with increasing power, while *RP on β_2* remains accurate as long as the accumulated dispersion is sufficiently small, as we will show.

In this paper, we extend the *first-order RP on β_2* of [5] to *dual-polarization* systems by deriving the perturbative solution of the Manakov equation. We benchmark the proposed model against dual-polarization RP on γ , demonstrating the operational regions where each approach excels. To the best of our knowledge, this is the first analysis and validation of dual-polarization RP on β_2 .

RP on β_2 for Dual-Polarization Systems

The propagation of dual-polarization signals through an optical fiber link is governed, in the absence of polarization mode dispersion, by the Manakov equation [7]

$$\frac{\partial \mathbf{A}(t, z)}{\partial z} = -j\frac{\beta_2}{2} \frac{\partial^2 \mathbf{A}(t, z)}{\partial t^2} + j\frac{8}{9} \gamma e^{-\alpha z} \|\mathbf{A}\|^2 \mathbf{A}(t, z), \quad (1)$$

where the vector $\mathbf{A}(t, z) = [A_x(t, z), A_y(t, z)]^T$ is the Jones vector representing the propagating two-polarization power-normalized optical field, β_2 is the GVD parameter, γ is the fiber Kerr coefficient, α is the attenuation coefficient, and $\|\mathbf{A}\|^2 = |A_x(t, z)|^2 + |A_y(t, z)|^2$ is the total instantaneous signal power.

The RP method [2] on the GVD parameter [5] expands the solution of (1) in powers of β_2 . We apply it to the dual-polarization signal $\mathbf{A}$ and truncate the power series to the first order, obtaining

$$\mathbf{A}_{\text{RP}}(t, z) = \mathbf{A}_0(t, z) + \beta_2 \mathbf{A}_1(t, z) \quad (2)$$

where $\mathbf{A}_0$ and $\mathbf{A}_1$ are determined in what follows. Substituting (2) in (1) and setting $\beta_2 = 0$ yields the zeroth-order RP on β_2 solution

$$\mathbf{A}_0(t, z) = \mathbf{A}(t, 0) e^{j\frac{8}{9} \gamma \|\mathbf{A}(t, 0)\|^2 G(z)}, \quad (3)$$

where $G(z) = (1 - e^{-\alpha z})/\alpha$ is the fiber effective length. This solution only includes a nonlinear phase rotation, depending on the overall input power $\|A(t, 0)\|^2$ of both polarizations, applied pointwise to the signal, it does not take into account the channel memory determined by chromatic dispersion.

Substituting (2) into (1) and grouping only the terms proportional to β_2 yields, after evaluating analytically the integrals that solve the resulting differential equation,

$$A_1(t, z) = B(t, z) e^{j \frac{8}{9} \gamma \|A(t, 0)\|^2 G(z)}, \quad (4)$$

where

$$B(t, z) = -M(t)z + R(t)G_1(z) + P(t)G_2(z) - j \frac{16}{9} \gamma A(t, 0) \operatorname{Re}\{\langle A^*(t, 0), V(t, z) \rangle\} \quad (5)$$

carries the interaction between GVD and nonlinear effects. Here $\langle u, v \rangle$ is the standard inner product for complex vector spaces, and

$$\begin{aligned} M(t) &= \frac{j}{2} \frac{d^2 A(t, 0)}{dt^2}, \\ R(t) &= \frac{4}{9} \gamma \frac{dH(t)}{dt} A(t, 0) + \frac{8}{9} \gamma H(t) \frac{dA(t, 0)}{dt}, \\ P(t) &= j \frac{32}{81} \gamma^2 H^2(t) A(t, 0), \end{aligned} \quad (6)$$

with $H(t) \triangleq d\|A(t, 0)\|^2/dt$ being the time derivative of the instantaneous input power. The function

$$\begin{aligned} V(t, z) &= M(t)[G(z)z - G_1(z)] \\ &\quad - R(t)[G(z)G_1(z) - G_2(z)] \\ &\quad - P(t)[G(z)G_2(z) - G_3(z)] \end{aligned} \quad (7)$$

couple the two polarizations through the inner product in the r.h.s. of (5), and, finally,

$$G_n(z) = \int_0^z G^n(z') dz' \quad (8)$$

are the effective lengths to the power of n integral functions, available in closed form [5, Eq. 27–29].

System Setup and Performance

We compare the first-order RP on β_2 in (2) (called RP- β_2) against the first-order RP on γ in their dual-polarization form [8] (called RP- γ). At the receiver, i.e., at $z = L$ (link length), the model error is measured, after applying chromatic dispersion compensation, by the normalized square deviation (NSD) defined as [2, Eq. 27]

$$\text{NSD} = \frac{\int_{-\infty}^{+\infty} \|A_{\text{SSFM}}(t, L) - A_{\text{RP}}(t, L)\|^2 dt}{\int_{-\infty}^{+\infty} \|A_{\text{SSFM}}(t, L)\|^2 dt}, \quad (9)$$

where A_{SSFM} represents the two continuous-time output waveforms of reference, computed using

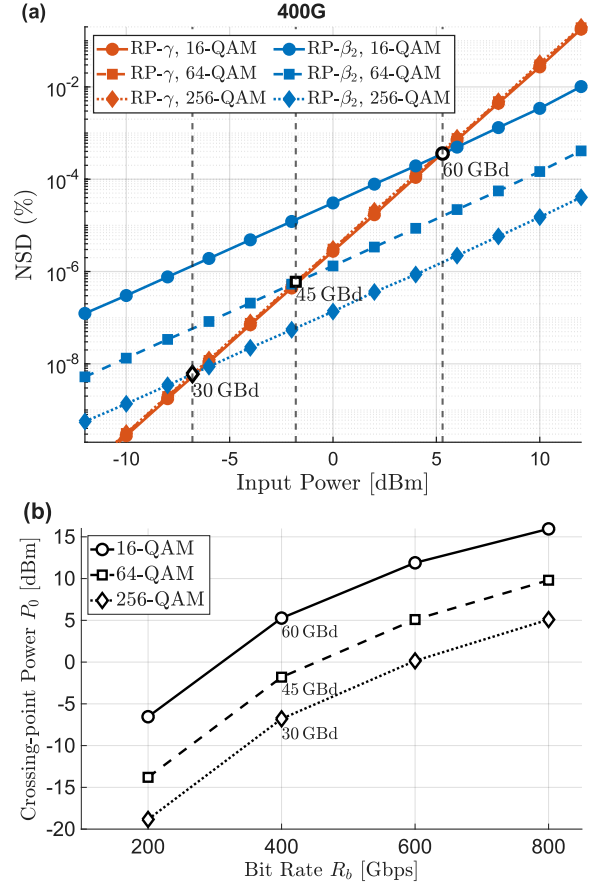


Fig. 1: (a) NSD vs. total input power for RP- β_2 (blue) and RP- γ (orange), at a fixed net bit rate of 400 Gbps; (b) Crossing-point power P_0 vs. net bit rate.

the symmetric SSFM with a sufficiently large number of steps. As a representative O-band DCI coherent link, we consider an SMF of $L = 20$ km, $\alpha = 0.4$ dB km⁻¹, $\gamma = 1.4$ W⁻¹ km⁻¹, $\beta_2 = -1$ ps² km⁻¹, $\beta_3 = 0.0765$ ps³ km⁻¹, dual-polarization transmission, and root-raised-cosine pulse shaping.

Fig. 1(a) depicts the NSD of the two RP models as a function of the total input power, at a fixed net bit rate of 400 Gbps and for three modulation formats: 16-, 64- and 256-QAM. Since net bit rate R_b is fixed, the M -QAM symbol rate $R_s = R_b / (2 \log_2 M \times 0.83)$ is decreased from 60 GBd for 16-QAM to 30 GBd for 256-QAM, where the factor 2 accounts for the polarizations and 0.83 accounts for a typical forward-error-correction overhead. We observe that the NSD of RP- γ grows steeply with power, and is virtually identical for the three modulation orders, showing that the model's accuracy is only sensitive to power but not to symbol rate. On the other hand, the NSD of RP- β_2 has a lower slope, and a different level for each modulation order, set by the dependence on symbol rate. The crossing point of each pair of curves occurs at an input power P_0 that decreases as the modulation order increases. No additional dependence on modulation order has been observed in the error between the RP models and SSFM.

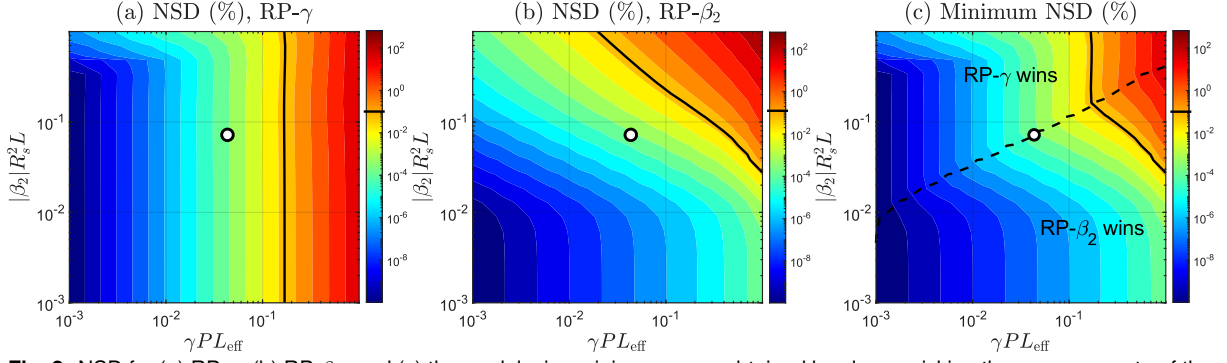


Fig. 2: NSD for (a) RP- γ , (b) RP- β_2 , and (c) the model-wise minimum error, obtained by always picking the more accurate of the two models. The solid black contour is the NSD = 0.1 % accuracy threshold, also marked on each colorbar. In panel (c) the dashed line is made by the crossing-points of equal NSD, partitioning the plane into the RP- γ and RP- β_2 favorable regions. The circle marker is the crossing-point of Fig. 1(a).

Fig. 1(a) shows that above P_0 , RP- β_2 is the more accurate model; below P_0 , RP- γ is preferred. To translate this observation into practical design guidelines, Fig. 1(b) shows the crossing-point power P_0 as a function of net bit rate, for the same O-band SMF and link length, with 16-, 64-, and 256-QAM. The value of P_0 increases with bit rate because a higher R_s enlarges the GVD-induced error of RP- β_2 , shifting the balance toward RP- γ . For a fixed bit rate, higher-order modulation formats require a lower R_s and therefore yield a lower P_0 , enlarging the region where RP- β_2 is advantageous. For instance, at 400 Gbps the crossing-point power moves from 5 dBm for 16-QAM down to -7 dBm for 256-QAM, which is well within the typical operating range of O-band DCI systems.

The crossing-point power and the model accuracy depend on all system parameters. To gain a unified view of the complementarity of the RP models, we introduce two dimensionless parameters:

$$\gamma PL_{\text{eff}}, \quad |\beta_2|R_s^2L, \quad (10)$$

where P is the total input power and $L_{\text{eff}} = G(L) = (1 - e^{-\alpha L})/\alpha$ is the effective length. The quantity γPL_{eff} is the nonlinear phase rotation argument and represents the magnitude of nonlinearities: it controls the error of RP- γ , whose zeroth-order solution neglects nonlinearities. The quantity $|\beta_2|R_s^2L$ represents accumulated chromatic dispersion at the end of the link, and it controls the error of RP- β_2 , whose zeroth-order solution neglects dispersion.

Fig. 2 reports the NSD for the first-order RP- γ , the first-order RP- β_2 , and the model-wise minimum error, obtained by always picking the model with the lower NSD. The contour plots are obtained by sweeping γ and $|\beta_2|$ at fixed $P = 5$ dBm, $R_s = 60$ GBd, $L = 20$ km, $\alpha = 0.4$ dB km $^{-1}$, for a dual-polarization 16-QAM format ($R_b = 400$ Gbps). On every panel, the solid black contour marks the NSD = 0.1 % threshold under which a model is empirically defined accurate (also indicated by

a horizontal black tick on each colorbar). The crossing-point power P_0 for the O-band 16-QAM scenario of Fig. 1(a) is marked on the plane with a circle marker.

While the NSD of RP- γ (see Fig. 2(a)) is nearly insensitive to $|\beta_2|R_s^2L$, the NSD of RP- β_2 (see Fig. 2(b)) grows more rapidly along $|\beta_2|R_s^2L$ than along γPL_{eff} (the color stripes are denser along $|\beta_2|R_s^2L$). This confirms the complementarity of the two approaches. Fig. 2(c) shows in further detail the complementarity of the two models: the dashed line consists of the crossing points of equal accuracy of the models and partitions the plane into a region where RP- γ is preferred (top-left) and a region where RP- β_2 is preferred (bottom-right). The overall 0.1 % accuracy region is the union of the two individual accurate regions and covers a large portion of the plane. In the top-right region, both first-order models are inaccurate, and thus, the choice of model does not matter.

Conclusion

We extended the first-order regular perturbation on β_2 to dual-polarization systems and validated it against SSFM for a wide range of operating scenarios, including short-reach O-band links for next-generation DCIs. For a 20 km-long link at 400 Gbps, RP- β_2 outperforms RP- γ at launch powers above approximately 5 dBm with 16-QAM, or lower when using higher-order modulation formats. Through the analysis in terms of the dimensionless parameters γPL_{eff} and $|\beta_2|R_s^2L$, we confirmed that the accuracy of RP- β_2 depends mainly on $|\beta_2|R_s^2L$, while RP- γ depends mostly on γPL_{eff} . Furthermore, we studied their complementarity by identifying regions of applicability for each model. By selecting the minimum NSD across the two models, 0.1 % NSD accuracy can be achieved over a vast region of the parameter space. This demonstrates that RP- β_2 , used alongside RP- γ , provides a computationally efficient analytical tool for dual-polarization short-reach systems. The application of RP- β_2 to DSP algorithms such as digital back-propagation will be investigated in future work.

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